

Pemrograman Web Lanjut

Tugas Login dan Captcha Proyek PBL

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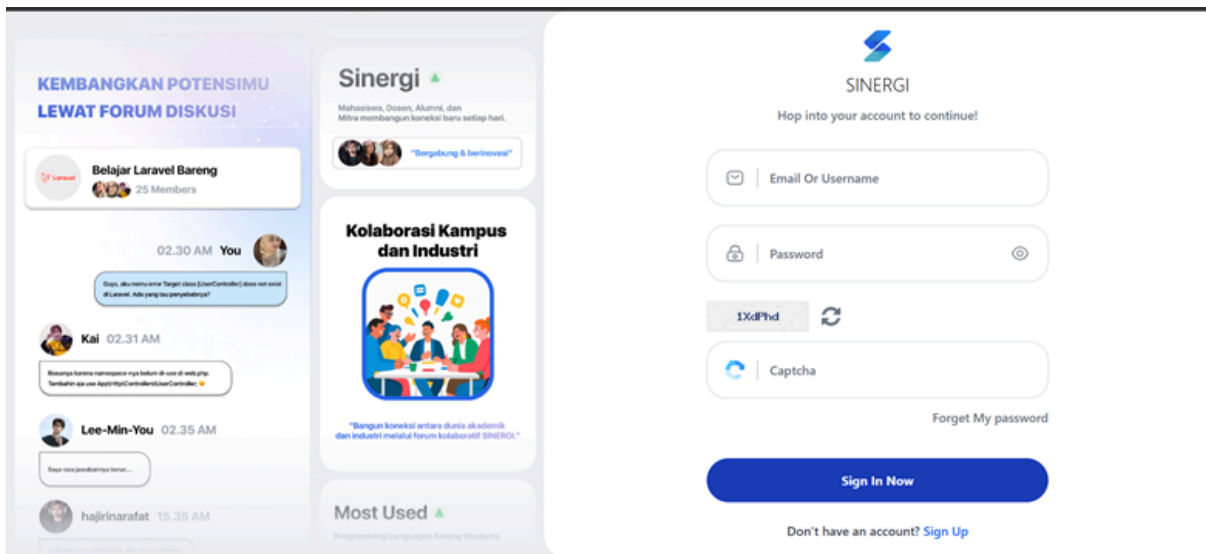
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TI 3 A

**PROGRAM STUDI TEKNIK INFORMATIKA
JURUSAN TEKNIK INFORMATIKA DAN KOMPUTER
POLITEKNIK NEGERI JAKARTA**

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1. Screenshot Hasil Captcha di Form Sign-in atau Login



2. Fungsi Generate dan Refresh Captcha

```
public function refreshCaptcha()
{
    $this->generateAndStoreCaptcha();
    header('Content-Type: application/json');
    echo json_encode(['status' => 'success']);
}

private function generateAndStoreCaptcha()
{
    $length = 6;

    $charset = "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz123456789";

    $captchaText = "";

    for ($i = 0, $n = strlen($charset); $i < $length; ++$i) {
        $captchaText .= $charset[random_int(0, $n - 1)];
    }

    $_SESSION['captcha'] = $captchaText;
}
```

3. Kode Fungsi Auth untuk Login dan Captcha

```
1 public function signInAction()
2 {
3     if ($SERVER['REQUEST_METHOD'] == 'POST') {
4         $identifier = $_POST['username_or_email'] ?? '';
5         $password = $_POST['password'] ?? '';
6         $captchaInput = $_POST['captcha'] ?? '';
7
8         if (empty($identifier) || empty($password) || empty($captchaInput)) {
9             $_SESSION['login_error'] = "Semua field wajib diisi!";
10             header('Location: ' . BASEURL . '/sign-in');
11             exit();
12         }
13
14         $sessionCaptcha = $_SESSION['captcha'] ?? '';
15
16         if (empty($sessionCaptcha) || $captchaInput != $sessionCaptcha) {
17             $_SESSION['login_error'] = "Captcha yang Anda masukkan salah!";
18             unset($_SESSION['captcha']);
19             header('Location: ' . BASEURL . '/sign-in');
20             exit();
21         }
22
23         unset($_SESSION['captcha']);
24
25         $user = $this->loginModel->getUserByUsernameOrEmail($identifier);
26
27         if ($user && password_verify($password, $user['PASSWORD'])) {
28
29             if ($user['ROLE'] == 'MAHASISWA') {
30                 $tahunMasuk = (int)$user['TAHUN_MASUK'];
31                 $jenjangStudi = $user['JENJANG_STUDI'];
32                 $userId = $user['ID'];
33
34                 if ($tahunMasuk > 0 && !empty($jenjangStudi)) {
35                     $durasiStudi = 0;
36                     if ($jenjangStudi == 'D4') {
37                         $durasiStudi = 4;
38                     } elseif ($jenjangStudi == 'D3') {
39                         $durasiStudi = 3;
40                     }
41
42                     if ($durasiStudi > 0) {
43                         $tahunLulus = $tahunMasuk + $durasiStudi;
44                         $bulanLulus = 10;
45                         $tahunSekarang = (int)date('Y');
46                         $bulanSekarang = (int)date('m');
47
48                         if ($tahunSekarang > $tahunLulus || ($tahunSekarang == $tahunLulus && $bulanSekarang >= $bulanLulus)) {
49                             $this->loginModel->updateUserRole($userId, 'ALUMNI');
50                             $user['ROLE'] = 'ALUMNI';
51                             $this->createSession($user);
52                             header('Location: ' . BASEURL . '/forums');
53                             exit();
54                         }
55                     }
56                 }
57             }
58
59             $this->createSession($user);
60             $role = $user['ROLE'];
61
62             if ($role == 'MAHASISWA' || $role == 'DOSEN') {
63                 header('Location: ' . BASEURL . '/homepage');
64                 exit();
65             } elseif ($role == 'ALUMNI' || $role == 'MITRA') {
66                 header('Location: ' . BASEURL . '/forums');
67                 exit();
68             } else {
69                 header('Location: ' . BASEURL . '/dashboard');
70                 exit();
71             }
72         } else {
73             $_SESSION['login_error'] = "Username, Email, atau Password salah!";
74             header('Location: ' . BASEURL . '/sign-in');
75             exit();
76         }
77     } else {
78         header('Location: ' . BASEURL . '/sign-in');
79         exit();
80     }
81 }
82
83 }
```

4. Query SQL Table Users

```
1 CREATE TABLE USERS (  
2   ID VARCHAR2(255) PRIMARY KEY,  
3   USERNAME VARCHAR2(255) UNIQUE NOT NULL,  
4   PERSONAL_NUMBER VARCHAR2(255),  
5   PRODI VARCHAR2(255),  
6   FULL_NAME VARCHAR2(255) NOT NULL,  
7   TAHUN_MASUK NUMBER(4),  
8   EMAIL VARCHAR2(255) UNIQUE NOT NULL,  
9   PASSWORD VARCHAR2(255) NOT NULL,  
10  PATH_PHOTO VARCHAR2(500),  
11  JENJANG_STUDI VARCHAR2(255),  
12  ROLE VARCHAR2(255) NOT NULL,  
13  CONSTRAINT CHK_USERS_ROLE CHECK (ROLE IN ('ADMIN', 'MAHASISWA', 'DOSEN', 'ALUMNI', 'MITRA'))  
14 );
```

5. Fungsi Membuat gambar captcha

```
1 <?php  
2 session_start();  
3  
4  
5 $captcha = $_SESSION['captcha'] ?? 'ERROR';  
6 $width = 130;  
7 $height = 40;  
8  
9 $image = imagecreate($width, $height);  
10  
11 $bg = imagecolorallocate($image, 240, 240, 240);  
12 $textColor = imagecolorallocate($image, 20, 40, 100);  
13  
14 for ($i = 0; $i < 50; $i++) {  
15     $noiseColor = imagecolorallocate($image, rand(150, 255), rand(150, 255), rand(150, 255));  
16     imagesetpixel($image, rand(0, $width), rand(0, $height), $noiseColor);  
17 }  
18  
19 $fontSize = 5;  
20 $x = (imagesx($image) - imagefontwidth($fontSize) * strlen($captcha)) / 2;  
21 $y = (imagesy($image) - imagefontheight($fontSize)) / 2;  
22 imagestring($image, $fontSize, $x, $y, $captcha, $textColor);  
23  
24 header('Content-Type: image/png');  
25 imagepng($image);  
26 imagedestroy($image);  
27
```

6. Fungsi menampilkan gambar Captcha dan Refresh Captcha

```
1 <div>
2   <div class="flex gap-3 items-center mb-3">
3     <img id="captcha-img" src="" /> <span>{{BASICURL}} {{captcha.php}} alt="captcha">
4     <button type="button" id="refresh-captcha" style="cursor:pointer;" class="p-1 rounded-md text-gray-600 hover:bg-gray-100 hover:text-gray-900">
5       
6     </button>
7   </div>
8   <div class="relative">
9     <div class="group relative">
10      <input type="text" id="captcha" name="captcha" class="w-full h-[28px] px-[8px] py-6 pt-5 pb-2 font-seebold text-gray-900 border-[1.5px] border-gray-300 rounded-[24px] focus:outline-none focus:border-blue-500 peer:transition-all placeholder=" " required />
11      <label for="captcha" class="absolute left-[8px] top-1/2 -translate-y-1/2 text-gray-500 font-medium peer-focus:top-4 peer-focus:text-in peer-placeholder-shown:top-1/2 peer-[&not;(placeholder-shown)]:top-4 peer-[&not;(placeholder-shown)]:text-in transition-all">Captcha /label>
12      
13    <div class="absolute left-[8px] top-1/2 -translate-y-1/2 w-[1.5px] h-8 bg-gray-300"></div>
14  </div>
15 </div>
16 <a href="" /> <span>{{BASICURL}} {{forgot-password}}</span> class="block text-end mt-3 font-medium text-gray-500 hover:text-blue-500 hover:underline">Forgot My password />
17 </div>
18
19 <script>
20   const refreshButton = document.getElementById('refresh-captcha');
21   const captchaImage = document.getElementById('captcha-img');
22
23   refreshButton.addEventListener('click', function() {
24     fetch('{{BASICURL}} {{refresh-captcha}}')
25       .then(response => response.json())
26       .then(data => {
27         if (data.status === 'success') {
28           captchaImage.src = '{{BASICURL}} {{captcha.php}}' + new Date().getTime();
29         }
30       })
31       .catch(error => {
32         console.error('Error refreshing captcha: ', error);
33       });
34   });
35 </script>
```